$t\bar{t}$ and $hh \to b\bar{b}\,\ell\nu\,q\bar{q}'$: samples, the ten angles, and $f_{t\bar{t}}$ by OSDE

Tae Min Hong

PITT PACC and Department of Physics and Astronomy, University of Pittsburgh

October 7, 2026

Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

Goal

- Generate $pp \to t\bar{t} \to b\bar{b}\,W^+W^- \to b\bar{b}\,\ell\nu\,q\bar{q}'$ at parton, particle (gen) and detector (reco) level
- Validate the samples and the ten angles $\mathbf{x} = \{Y, \Theta, \boldsymbol{\theta}, \tilde{\boldsymbol{\theta}}\}$ used for the $t\bar{t}$ density $f_{t\bar{t}}(\mathbf{x})$
- Everything is scripted: `generate_samples.py`, `make_plots.py`, `make_slides.py`

Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

Sample generation

- MadGraph5_aMC@NLO 3.6.7 (LO, `sm`, 4-flavour proton) → Pythia8 → Delphes 3.5.1, ATLAS card, **no pileup**
- Decay chains keep spin correlations; no generator-level cuts:

```
generate    p p > t t~, (t > b w+, w+ > lep+ nul), (t~ > b~ w-, w- > q q)
add process p p > t t~, (t > b w+, w+ > q q), (t~ > b~ w-, w- > lep- nul~)
```

- $\sqrt{s} = 13$ TeV, $\ell = e, \mu$, 10000 events, seed 12345
- $\sigma = 136.5$ pb $\approx \sigma^{\mathrm{LO}}_{t\bar{t}} \times 2\,\mathcal{B}(W \to \ell\nu)\,\mathcal{B}(W \to q\bar{q}')$ (with $\ell = e, \mu$)

Object definitions and levels

	parton	parton + ℓ trigger	particle level (gen)	reco (Delphes ATLAS)
source	MG5 LHE, before shower	MG5 LHE, before shower	Pythia8 final state	Delphes ATLAS card
leptons	LHE e/μ, no cut	$p_T > 25$, $\|\eta\| < 2.47\,(e), 2.5\,(\mu)$	status-1 e/μ from $W/Z/h$ (τ)	Electron, Muon (same cuts)
jets	LHE $q, \bar{q}'$ (no cut)	LHE $q, \bar{q}'$ (no cut)	GenJet anti-k_t 0.4, $p_T > 25$, $\|\eta\| < 2.5$	Jet (same cuts)
b	LHE $b, \bar{b}$	LHE $b, \bar{b}$	$\Delta R(j, b) < 0.4$, $p_T^b > 5$	BTag bit 0
overlap	–	–	drop jets $\Delta R(j, \ell) < 0.4$	UniqueObjectFinder
E_T^{miss}	p_T^ν (LHE)	p_T^ν (LHE)	GenMissingET: $\sum$ all ν	MissingET: $-\sum$ calo towers
selection	none	lepton only	1ℓ, $\geq 4j$, $\geq 1b$, χ^2	1ℓ, $\geq 4j$, $\geq 1b$, χ^2
events kept	100%	71.7%	41.3%	24.4%

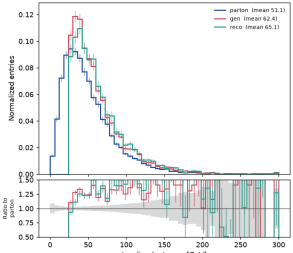
Parton level means the MG5 partons with *no* cuts; "parton + ℓ trigger" adds only the offline single-lepton cuts. Reco E_T^{miss} is the negative vector sum of the calorimeter towers; checked: muon and electron events show the same reco$-$gen E_T^{miss} along the lepton (-5.9 vs -5.2 GeV), so muons are not missed.

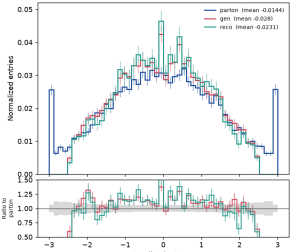
Cutflow

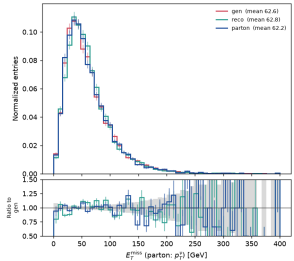
selection	gen		reco	
All events	10000	100.0%	10000	100.0%
exactly 1 lepton	7071	70.7%	4907	49.1%
≥ 4 jets	4196	42.0%	2830	28.3%
≥ 1 b-jet	4161	41.6%	2454	24.5%
≥ 2 b-jets	3305	33.0%	1221	12.2%
χ^2 reconstruction (from ≥ 1 b)	4134	41.3%	2440	24.4%

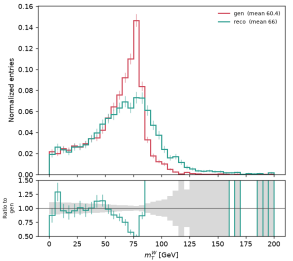
Fractions relative to all generated events.

Leptons and E_T^{miss}

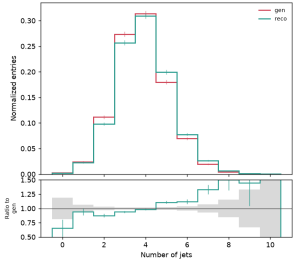


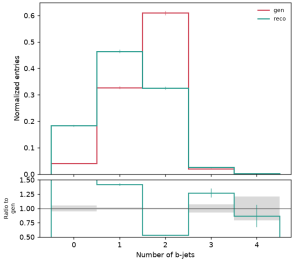


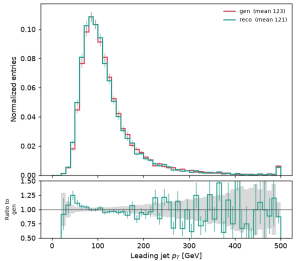


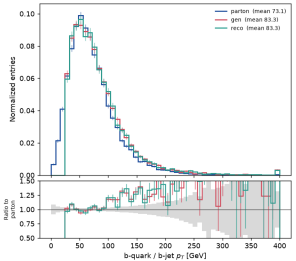


Jets


gen
reco
Normalised entries
Ratio to gen
Number of jets

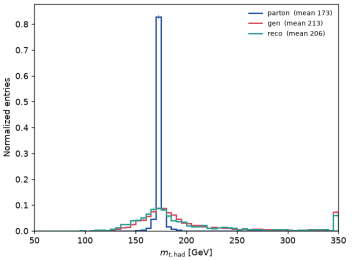

gen
reco
Normalised entries
Ratio to gen
Number of b-jets

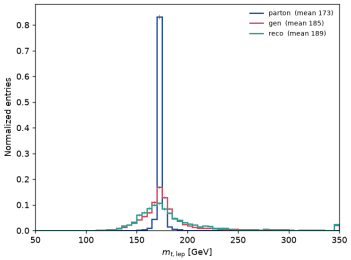

gen (mean 123)
reco (mean 121)
Normalised entries
Ratio to gen
Leading jet p_T [GeV]

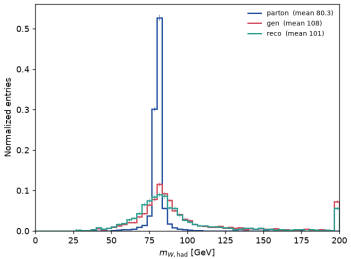

parton (mean 73.1)
gen (mean 83.3)
reco (mean 83.3)
Normalised entries
Ratio to parton
b-quark / b-jet p_T [GeV]

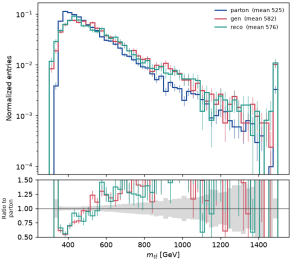
- $\chi^2 = \left(\frac{m_{jj}-m_W}{\sigma_W}\right)^2 + \left(\frac{m_{jjb}-m_t}{\sigma_t}\right)^2 + \left(\frac{m_{\ell\nu b}-m_t}{\sigma_t}\right)^2$ over assignments of the 6 leading jets
- b slots hold $\min(n_b, 2)$ b-tagged jets
- p_z^ν from the m_W constraint; both solutions enter the χ^2, so the m_t term resolves the two-fold ambiguity
- All four jets correct in 23% (gen) and 18% (reco) of reconstructed events

Reconstructed masses









Mass resolution (median $\pm$ 68% half-width, GeV)

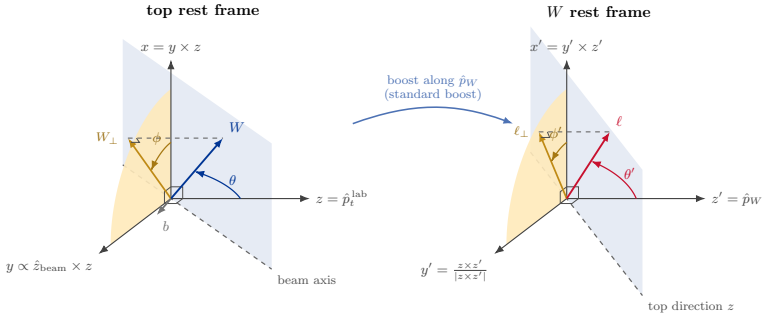
	$m_{t,\mathrm{had}}$, all	$m_{t,\mathrm{had}}$, correct jets	$m_{W,\mathrm{had}}$, all	$m_{W,\mathrm{had}}$, correct jets
gen	182 ± 47	175 ± 14	85 ± 29	83 ± 9
reco	177 ± 44	170 ± 16	84 ± 25	80 ± 10

The broad inclusive peaks come from wrong jet choices (combinatorics) and quarks that are out of acceptance, not from the jet response.

Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

Definition of the ten angles



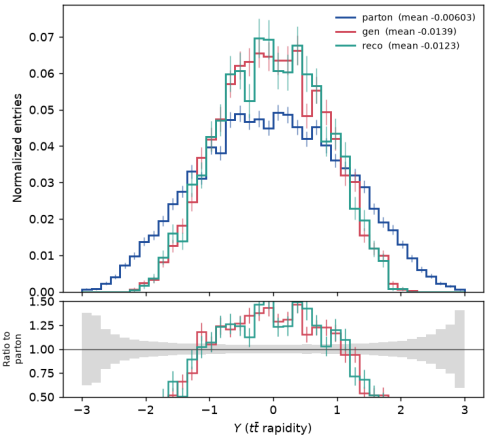
x–z plane (contains the beam and $\hat{p}_t^{\text{lab}}$; right: the x'–z' plane, contains $\hat{p}_W$ and the top direction) xy plane, perpendicular to z

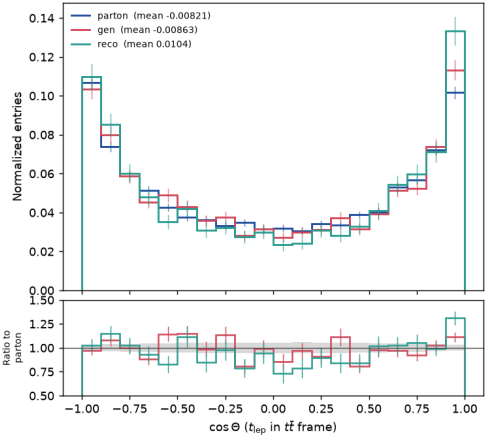
θ: angle between z and the W. ϕ: **azimuth of the** W **around** z: project the W onto the xy plane ($W_\perp$); ϕ is the angle from x to $W_\perp$, toward y.
(θ', ϕ'): the same for the charged lepton in the W rest frame, with (x', y', z'): $z' = \hat{p}_W$, $y' = z \times z'/|z \times z'|$ (in the xy plane), $x' = y' \times z'$, i.e. $(x', y', z') = R(\phi, \theta, 0)\,(x, y, z)$.
The beam is not along z (z is the top direction); it lies in the x–z plane with a negative x component, since $y \propto \hat{z}_{\text{beam}} \times z$.
Hadronic top: same construction with the down-type quark in place of ℓ.

The ten angles (as in the paper)

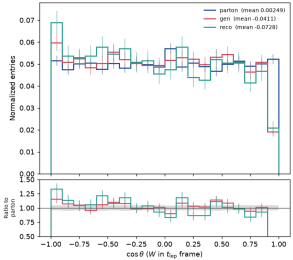
- Y: $t\bar{t}$ rapidity (lab); Θ: polar angle of t_{lep} in the $t\bar{t}$ frame
- (θ, ϕ): W in the top rest frame; $z = \hat{p}_t^{\text{lab}}$, $y = \hat{z}_{\text{beam}} \times z$, $x = y \times z$
- (θ', ϕ'): fermion in the W rest frame (boost from top frame); $z' = \hat{p}_W$, $y' = z \times z'$, $x' = y' \times z'$, i.e. $D(\phi, \theta, 0)$
- Leptonic top: fermion $= \ell$. Hadronic top ($\sim$): fermion = down-type quark
- gen/reco: down-type jet chosen by truth-matching to the LHE quarks (assumes up/down tagging, as in the paper)
- Check: building momenta from known angles and recovering them agrees to 10^{-14}

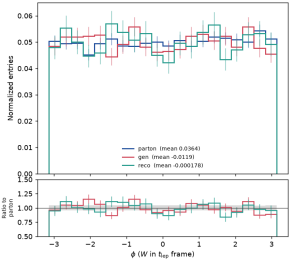
Production angles Y, $\cos\Theta$

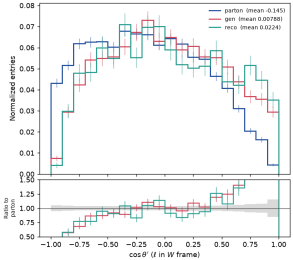


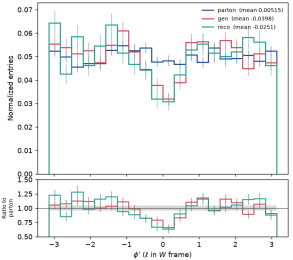


Leptonic top: $\theta, \phi, \theta', \phi'$

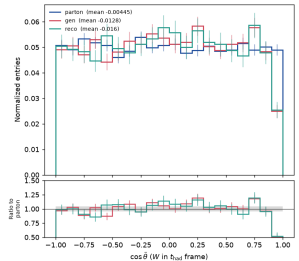


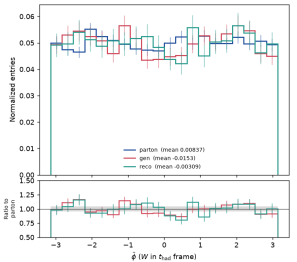


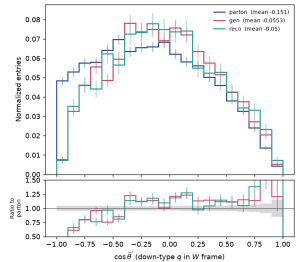


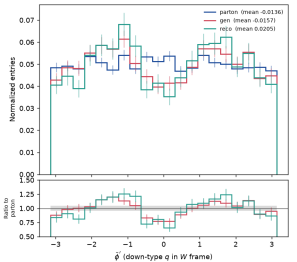


Hadronic top: $\tilde{\theta}, \tilde{\phi}, \tilde{\theta}', \tilde{\phi}'$









W helicity fractions from $\cos\theta'$ (moments)

	F_0	F_L	F_R
parton, ℓ	0.715 ± 0.013	0.288 ± 0.008	-0.003 ± 0.008
parton, down-type q	0.705 ± 0.013	0.298 ± 0.008	-0.003 ± 0.008
gen, ℓ	0.769 ± 0.019	0.107 ± 0.012	0.123 ± 0.012
gen, down-type q	0.954 ± 0.017	0.078 ± 0.011	-0.032 ± 0.011
reco, ℓ	0.687 ± 0.026	0.134 ± 0.017	0.179 ± 0.017
reco, down-type q	0.929 ± 0.023	0.086 ± 0.015	-0.014 ± 0.015

SM (LO): $F_0 = m_t^2/(m_t^2 + 2m_W^2) \approx 0.70$, $F_L \approx 0.30$, $F_R \approx 0$. Parton level reproduces it; gen/reco are distorted by acceptance and combinatorics (no unfolding).

Angle resolution vs parton level (correlation / median $|\Delta|$)

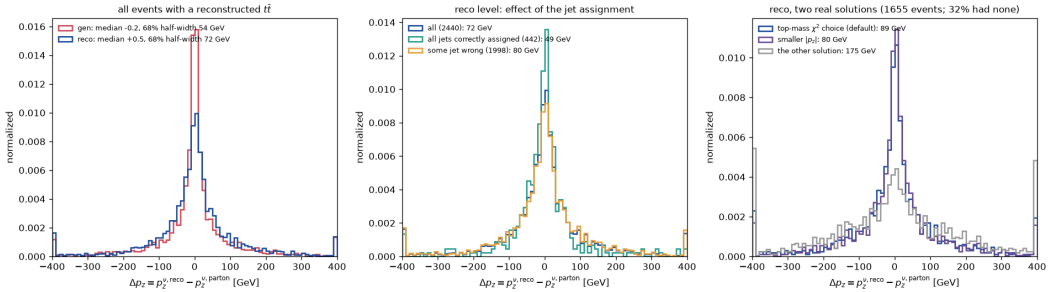
angle	gen, all	gen, correct jets	reco, all	reco, correct jets
Y	0.90 / 0.16	0.99 / 0.05	0.89 / 0.18	0.98 / 0.06
$\cos\Theta$	0.67 / 0.13	0.97 / 0.04	0.64 / 0.15	0.97 / 0.06
$\cos\theta$	0.70 / 0.14	0.89 / 0.06	0.62 / 0.22	0.84 / 0.11
ϕ	0.48 / 0.36	0.82 / 0.12	0.40 / 0.52	0.89 / 0.16
$\cos\theta'$	0.68 / 0.11	0.92 / 0.04	0.60 / 0.20	0.89 / 0.11
ϕ'	0.45 / 0.37	0.68 / 0.13	0.30 / 0.63	0.47 / 0.28
$\cos\tilde{\theta}$	0.59 / 0.18	0.98 / 0.04	0.45 / 0.24	0.97 / 0.05
$\tilde{\phi}$	0.40 / 0.46	0.88 / 0.09	0.34 / 0.69	0.90 / 0.09
$\cos\tilde{\theta}'$	0.41 / 0.18	0.97 / 0.04	0.31 / 0.25	0.96 / 0.06
$\tilde{\phi}'$	0.39 / 0.55	0.93 / 0.06	0.33 / 0.73	0.94 / 0.07

Neutrino p_z resolution

	median Δp_z [GeV]	68% half-width [GeV]
gen	-0.25	54
reco	0.5	72.2

$\Delta p_z = p_z^{\nu,\mathrm{reco}} - p_z^{\nu,\mathrm{parton}}$, events with a reconstructed $t\bar{t}$ system.

Neutrino Δp_z



Correct jets: 49 GeV vs 72 GeV for all events (68% half-width). No real solution in 32% of events. With two solutions, the top-mass χ^2 picks the one closer to the truth in 63% of events vs 65% for smaller $|p_z|$.

Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

- MadGraph5 built-in `heft` model (LO, $m_t \to \infty$), same Pythia8 + Delphes ATLAS chain

```
generate    g g > h h, (h > b b~), (h > lep+ nul q q)
add process g g > h h, (h > b b~), (h > lep- nul~ q q)
```

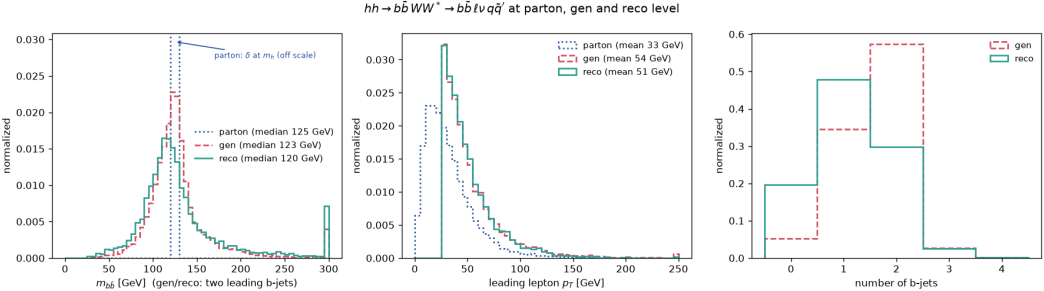
- $h \to WW^*$ has one off-shell W, so $h \to \ell\nu q\bar{q}'$ is a 4-body decay
- 20000 events, seed 2001; $\sigma \times \mathcal{B} = 0.112$ fb (LO HEFT with MG defaults; *not* an SM prediction, so plots use unit area)
- For tt-hypothesis quantities, hh uses the same χ^2; at parton level the b pairing is the one closest to m_t
- Prompt leptons: status-1 e/μ from W, Z, h, or a τ from one of these (in $h \to WW^*$ the lepton can come directly from the h)

Cutflow: $t\bar{t}$ vs hh (reco)

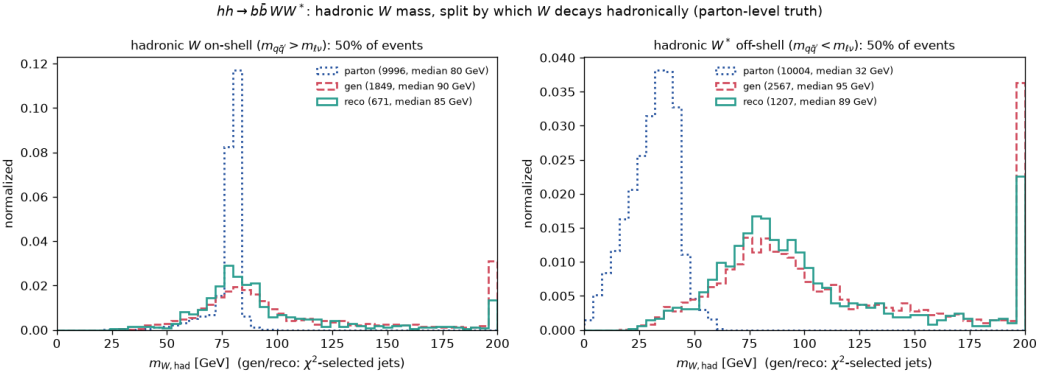
selection	$t\bar{t}$		hh	
All events	10000	100.0%	20000	100.0%
exactly 1 lepton	4907	49.1%	5403	27.0%
≥ 4 jets	2830	28.3%	2133	10.7%
≥ 1 b-jet	2454	24.5%	1887	9.4%
≥ 2 b-jets	1221	12.2%	920	4.6%
χ^2 reconstruction (from ≥ 1 b)	2440	24.4%	1878	9.4%

hh leptons are softer (from W^*) and less isolated, so the efficiency is lower.

hh sample: parton, gen and reco level



hh: hadronic W mass, on-shell W vs W^*



Hadronic boson is the W^* in 50% of events (true median 32 GeV), but the $t\bar{t}$-hypothesis χ^2 picks jet pairs near m_W (median 95 / 89 GeV at gen / reco): the reconstruction makes hh look like $t\bar{t}$.

Outline

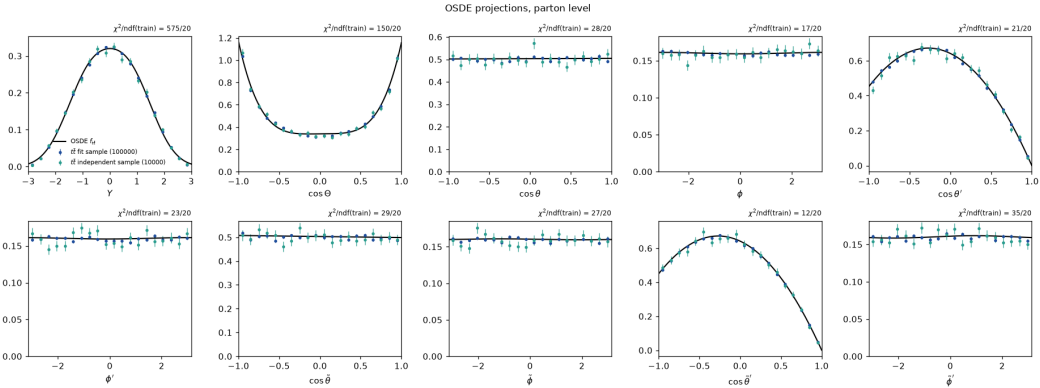
1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

Basis functions (as in the paper)

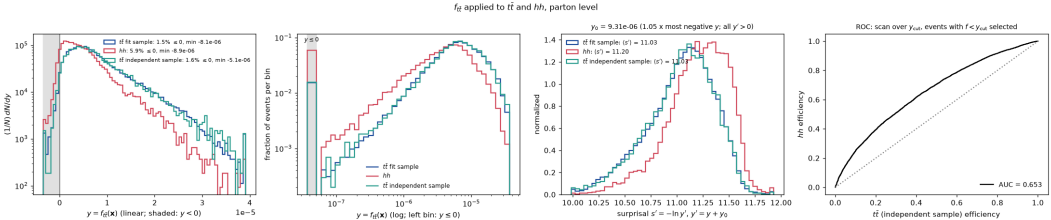
- $M^{j_1 j_2}_{m' m}(\phi, \theta, \phi', \theta') = \frac{\sqrt{2j_1+1}\sqrt{2j_2+1}}{4\pi} D^{j_1}_{m' m}(\phi, \theta, 0)\, D^{j_2}_{m 0}(\phi', \theta', 0)$, $j_1 \leq 1$, $j_2 \leq 2$: 24 per top
- $p_l(\cos\Theta) = \sqrt{\frac{2l+1}{2}} P_l$, $h_n(Y) = (2^n n! \sqrt{\pi})^{-1/2} H_n(Y) e^{-Y^2/2}$, $l < 5$, $n < 5$
- $B = M_{\mathbf{j}} M_{\tilde{\mathbf{j}}}\, p_l\, h_n$: $24 \times 24 \times 5 \times 5 = 14400$ coefficients
- OSDE: $\hat{c} = \frac{1}{N} \sum_k B^*(\mathbf{x}_k)$, with standard error; $f_{t\bar{t}} = \sum \hat{c}\, B$ (real, since the set is closed under conjugation)
- Hermite *functions*: bare H_n are orthogonal only with weight e^{-Y^2}
- **Three separate fits**: $f^{\text{parton}}_{t\bar{t}}$ from parton-level $t\bar{t}$ angles, applied to parton-level $t\bar{t}$ and hh; likewise f^{gen} (gen on gen) and f^{reco} (reco on reco)

Checks of the basis and of OSDE

- Unit tests: 12 passed
- Wigner $d^{j}_{m'm}$ agrees with closed forms and with `sympy` to 10^{-12}
- Gram matrices of M (exact quadrature), p_l, h_n equal the identity to 10^{-12}
- Toy: 2×10^5 events from a density with known coefficients; all 14400 recovered with pulls of unit width; projections and $\int f = 1$ reproduced
- Fitted f: imaginary part $< 10^{-10}$ of the real part
- Extended basis: Wigner d stable to $j = 24$ (Jacobi form); blocks disjoint and orthonormal; reduces exactly to the paper basis; leave-one-out equals brute-force refits

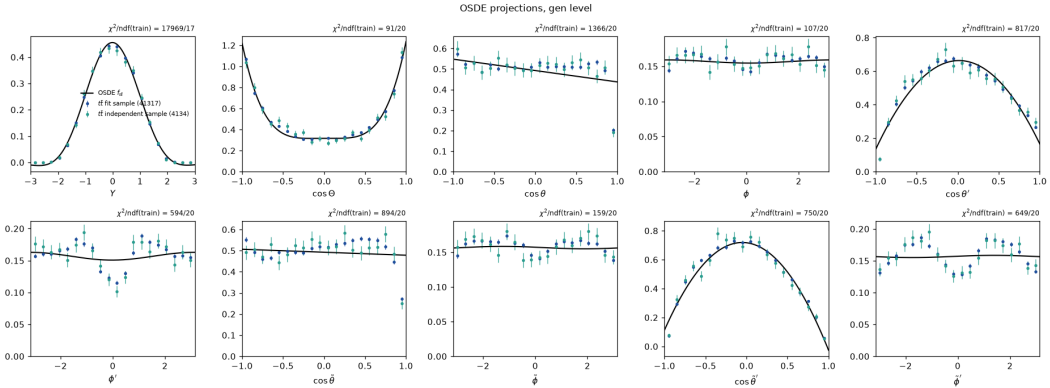


Black: $f_{t\bar{t}}$ integrated over the other nine angles. Largest χ^2/ndf vs training data: Y (575/20).



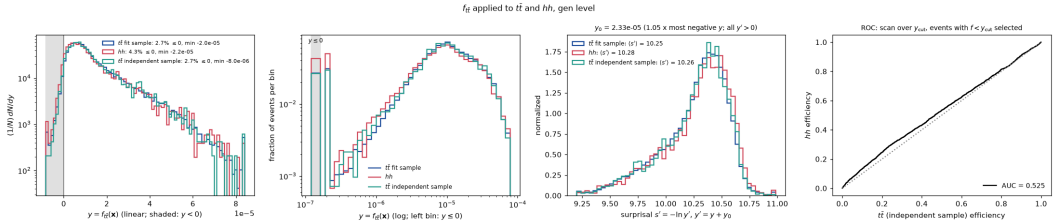
Left: $y = f_{t\bar{t}}(\mathbf{x})$; middle: surprisal $s = -\ln y$; right: hh vs $t\bar{t}$ efficiency when calling $f < y_{\text{cut}}$ anomalous.

$f_{t\bar{t}}^{\text{gen}}$ (fit on gen-level $t\bar{t}$, 41317 events): projections



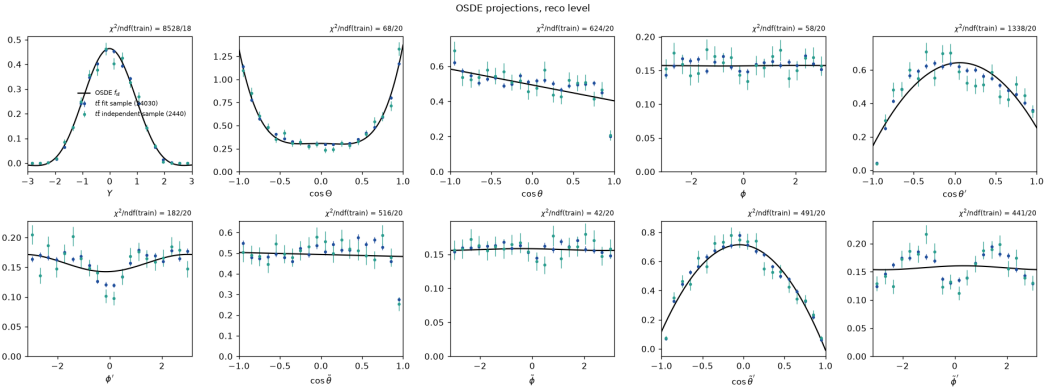
Black: $f_{t\bar{t}}$ integrated over the other nine angles. Largest χ^2/ndf vs training data: Y (17969/17).

$f_{t\bar{t}}^{\text{gen}}$ applied to gen-level $t\bar{t}$ (independent) and hh



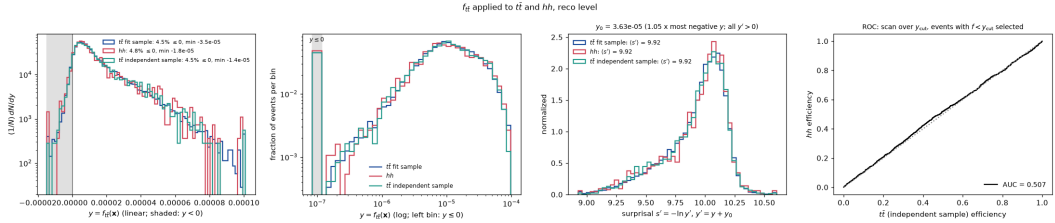
Left: $y = f_{t\bar{t}}(\mathbf{x})$; middle: surprisal $s = -\ln y$; right: hh vs $t\bar{t}$ efficiency when calling $f < y_{\text{cut}}$ anomalous.

$f_{t\bar{t}}^{\text{reco}}$ (fit on reco-level $t\bar{t}$, 24030 events): projections



Black: $f_{t\bar{t}}$ integrated over the other nine angles. Largest χ^2/ndf vs training data: Y (8528/18).

$f_{t\bar{t}}^{\text{reco}}$ applied to reco-level $t\bar{t}$ (independent) and hh



Left: $y = f_{t\bar{t}}(\mathbf{x})$; middle: surprisal $s = -\ln y$; right: hh vs $t\bar{t}$ efficiency when calling $f < y_{\text{cut}}$ anomalous.

$f_{t\bar{t}}$ summary

level	N_{train}	$\lvert\hat{c}\rvert > 3\sigma$	$\int f$	$f \leq 0$: $t\bar{t}/hh$	$\langle s \rangle$: $t\bar{t}/hh$	AUC
parton	100000	186	1.006	1.6% / 5.9%	12.07 / 12.56	0.653
gen	41317	330	0.985	2.7% / 4.3%	11.59 / 11.66	0.525
reco	24030	253	0.987	4.5% / 4.8%	11.53 / 11.56	0.507

$t\bar{t}$ numbers are from the independent test sample. $f \leq 0$ comes from the truncated series (Gibbs overshoot), as noted in the paper. AUC = probability that an hh event has lower $f_{t\bar{t}}$ than a $t\bar{t}$ event.

Fit sample vs independent sample: self-inclusion bias

level	self-term / f	$f \leq 0$: fit / fit LOO / indep.	$\langle s \rangle$: fit / fit LOO / indep.	KS vs indep.: fit / LOO
parton	7%	0.6% / 1.5% / 1.6%	11.98 / 12.08 / 12.07	0.041 / 0.010
gen	10%	1.0% / 2.7% / 2.7%	11.44 / 11.57 / 11.59	0.071 / 0.012
reco	16%	1.2% / 4.5% / 4.5%	11.33 / 11.53 / 11.53	0.103 / 0.012

Each fit event adds $\frac{1}{N}\sum_j |B_j(\mathbf{x}_k)|^2$ to its own f. Removing it (leave-one-out, LOO) makes the fit sample agree with the independent $t\bar{t}$ sample: the difference was entirely this bias, largest at reco (smallest N). Plots use LOO for the fit sample.

Diagnostics: statistical noise of the 14400 coefficients

level	all (paper)	$\lvert\hat{c}\rvert > 2\sigma$	$\lvert\hat{c}\rvert > 3\sigma$	$\lvert\hat{c}\rvert > 5\sigma$
parton	14400 / 0.653 / $+0.089$	751 / 0.663 / $+0.014$	186 / 0.666 / $+0.001$	54 / 0.669 / -0.005
gen	14400 / 0.525 / $+0.144$	827 / 0.525 / $+0.042$	330 / 0.525 / $+0.025$	153 / 0.527 / $+0.009$
reco	14400 / 0.507 / $+0.202$	712 / 0.507 / $+0.040$	253 / 0.504 / $+0.025$	127 / 0.507 / $+0.015$

Entries: coefficients kept / AUC(hh vs $t\bar{t}$ test) / $\langle s \rangle_{\text{test}} - \langle s \rangle_{\text{train}}$ (overfitting). Dropping insignificant terms removes the overfitting, but the gen/reco AUC stays ≈ 0.5, so the loss of separation is not caused by noise.

Diagnostics: Hermite scale for Y (χ^2 of the Y projection, train)

level	$s = 1.0$	$s = 0.9$	$s = 0.8$	$s = 0.7$	$s = \mathrm{std}(Y)$
parton	660/24	998/24	379/24	507/24	230/24
gen	2298/20	2956/20	548/20	728/20	548/20
reco	4485/21	3864/21	721/21	361/21	515/21

The paper uses $s = 1$ (default here); `osde_fit.py --y-scale` changes it. Gen/reco Y is narrower (std ≈ 0.8), so $s = 1$ fits poorly there and f overshoots below zero at large $|Y|$.

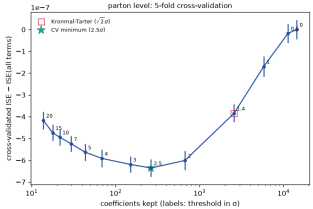
Outline

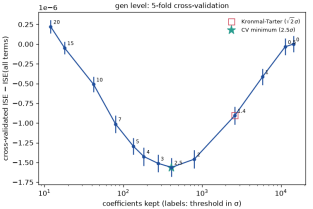
1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

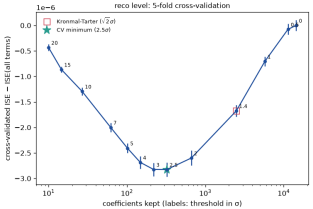
Criterion for dropping coefficients

- Keep a term iff $|\hat{c}|/\sigma(\hat{c}) > t$ (constant always kept). Conjugate partners have equal $|\hat{c}|$ and σ, so they are kept together and f stays real (checked)
- Kronmal–Tarter ($t = \sqrt{2}$): dropping a term costs $|c|^2$, keeping it costs σ^2; keep iff $|\hat{c}|^2 > 2\sigma^2$ (minimizes the integrated squared error per term)
- With 14400 mostly-null terms, $\sqrt{2}\sigma$ still admits $\sim$16% of the pure-noise terms
- Choose t by 5-fold cross-validation: for an orthonormal basis, $\mathrm{ISE} - \int f^2 = \sum_{\text{kept}} |\hat{c}|^2 - 2\,\mathrm{Re} \sum_{\text{kept}} \hat{c}\, \hat{c}^*_{\text{held-out}}$ (no integrals needed)
- CV minimum at $t = 2.5$: 326 (parton), 459 (gen), 342 (reco) of 14400 terms; flat out to 5σ: 54, 153, 127

Cross-validated error vs number of coefficients







Simplified $f_{t\bar{t}}$: performance

level	terms kept	number	AUC	$f \leq 0$ ($t\bar{t}$ indep.)	$f \leq 0$ (hh)	$\langle s \rangle$ fit LOO / indep.
parton	all (paper)	14400	0.653	1.6%	5.9%	12.08 / 12.07
	$\sqrt{2}\sigma$ (K–T)	2629	0.658	1.0%	4.8%	12.04 / 12.06
	CV best	326	0.665	0.2%	2.4%	12.06 / 12.06
	5σ	54	0.669	0.1%	2.1%	12.07 / 12.06
gen	all (paper)	14400	0.525	2.7%	4.3%	11.57 / 11.59
	$\sqrt{2}\sigma$ (K–T)	2725	0.526	1.9%	2.9%	11.53 / 11.57
	CV best	459	0.525	0.9%	1.3%	11.56 / 11.58
	5σ	153	0.527	1.0%	1.5%	11.57 / 11.58
reco	all (paper)	14400	0.507	4.5%	4.8%	11.53 / 11.53
	$\sqrt{2}\sigma$ (K–T)	2551	0.508	2.7%	2.8%	11.47 / 11.52
	CV best	342	0.505	1.1%	1.5%	11.52 / 11.54
	5σ	127	0.507	1.5%	1.9%	11.54 / 11.55

Dropping insignificant terms cuts the negative-f (Gibbs) fraction and keeps the hh separation: the simplified f is as good, with $\sim$1–3% of the terms.

Leading terms at parton level (after the constant)

#	term	$\lvert\hat{c}\rvert/\sigma$	physics
1	$P_2(\cos\Theta)$	115	production angle
2	$M_{0,0}^{01}$	88	W helicity (lep)
3	$\tilde{M}_{0,0}^{01}$	87	W helicity (had)
4	$M_{0,0}^{02}$	80	W helicity (lep)
5	$\tilde{M}_{0,0}^{02}$	79	W helicity (had)
6	$h_2(Y)$	59	rapidity
7	$P_4(\cos\Theta)$	34	production angle
8	$M_{0,0}^{01} \times P_2(\cos\Theta)$	32	W helicity (lep); production angle

M: leptonic top, $\tilde{M}$: hadronic top; $j_1 = 0$: W helicity, $j_1 = 1$: top spin. Production (P_l), W helicity and Y dominate.

What the significant ($> 5\sigma$) terms describe

physics content of the term	parton	gen	reco
production angle + top pol. $\times W$ hel. interference	–	30	26
rapidity + top pol. $\times W$ hel. interference	–	24	12
top pol. $\times W$ hel. interference + $t\bar{t}$ spin correlation	16	–	–
W helicity + production angle	12	8	6
production angle + rapidity + top polarization	–	12	12
W helicity + production angle + rapidity	1	11	11
production angle + rapidity + top pol. $\times W$ hel. interference	–	11	11
W helicity	8	6	6
production angle + rapidity	2	7	6
top pol. $\times W$ hel. interference	–	7	7
total	**53**	**152**	**126**

Parton: production, W helicity and $t\bar{t}$ spin correlations; imaginary parts consistent with 0 (max 1.4σ), as expected without CP violation. Reco: spin correlations no longer resolved; acceptance-induced "polarization" terms appear instead.

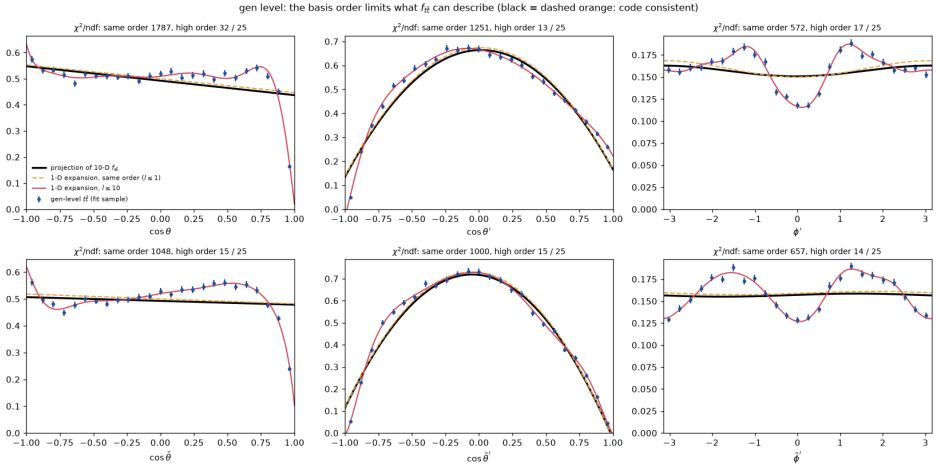
Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

Why f^{gen}, f^{reco} miss the data with the paper basis

- $j_1 \leq 1$: at most *linear* in $\cos\theta$; $j_2 \leq 2$: at most *quadratic* in $\cos\theta'$; $|m| \leq 1$: only $\cos\phi'$, $\sin\phi'$
- Complete for the physical (parton) decay distribution, but the acceptance cuts create sharp edges ($\cos\theta \to 1$, $\cos\theta' \to -1$, $\phi' \approx 0$) that need higher orders
- Code check: the projection of f equals a direct 1-D expansion at the same order (max difference 1.9%)
- At gen level, same-order χ^2: 1787, 1251, 572, 1048, 1000, 657 (25 bins); higher order: 32, 13, 17, 15, 15, 14

Paper basis at gen level: the order, not the fit, limits f



How many orders each angle needs (1-D cross-validated order)

angle	parton	gen	reco
$\cos\theta$	0	13	18
ϕ	0	4	4
$\cos\theta'$	2	6	5
ϕ'	1	12	4
$\cos\tilde{\theta}$	1	7	7
$\tilde{\phi}$	0	4	3
$\cos\tilde{\theta}'$	2	8	24
$\tilde{\phi}'$	1	4	5
$\cos\Theta$	10	6	13
Y	10	8	24
needs	$j_1 \geq 1, j_2 \geq 2$	$j_1 \geq 13, j_2 \geq 12$	$j_1 \geq 18, j_2 \geq 24$

Order = Legendre degree ($\cos$), harmonic (ϕ), Hermite order (Y, scale = std(Y)). Parton level confirms the paper: $j_1 \leq 1$, $j_2 \leq 2$ (only $\cos\Theta$, Y need more). A full product at gen/reco orders would need $> 10^6$ terms per top pair.

Extended (block) basis

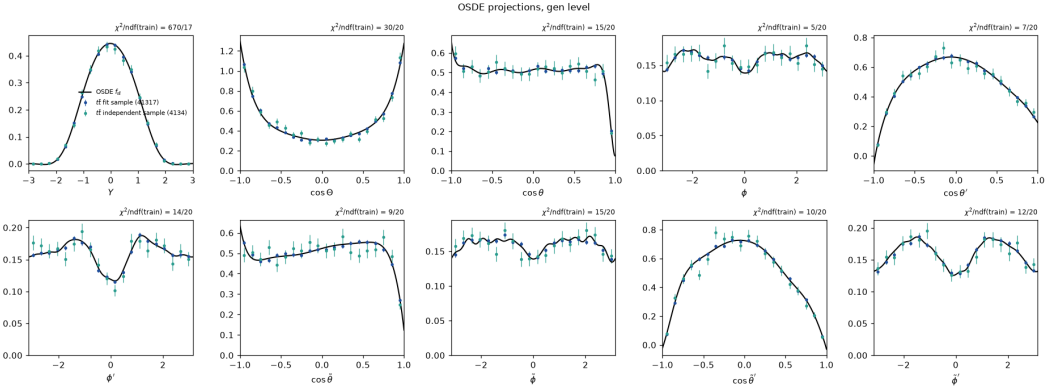
- Union of disjoint subsets of the full orthonormal product basis $\Rightarrow$ still orthonormal, OSDE unchanged
- A: the paper's full product (all correlations at physics order); L, H: high-order single-top blocks (all (θ, ϕ) terms, all $m' = 0$ terms) with the other top constant; P, Y: high-order $\cos\Theta$ and Y
- Orders from the scan, capped at 16: parton: $j_1 \leq 1$, $j_2 \leq 2$, 11 Legendre, 11 Hermite ($s = 1.111$); gen: $j_1 \leq 13$, $j_2 \leq 12$, 7 Legendre, 9 Hermite ($s = 0.799$); reco: $j_1 \leq 16$, $j_2 \leq 16$, 14 Legendre, 17 Hermite ($s = 0.791$)
- Terms: parton 14436, gen 32494, reco 49867; then the same cross-validated truncation
- Trade-off: correlations *between* the two tops stay at the physics order

Projection χ^2/ndf vs the $t\bar{t}$ fit sample

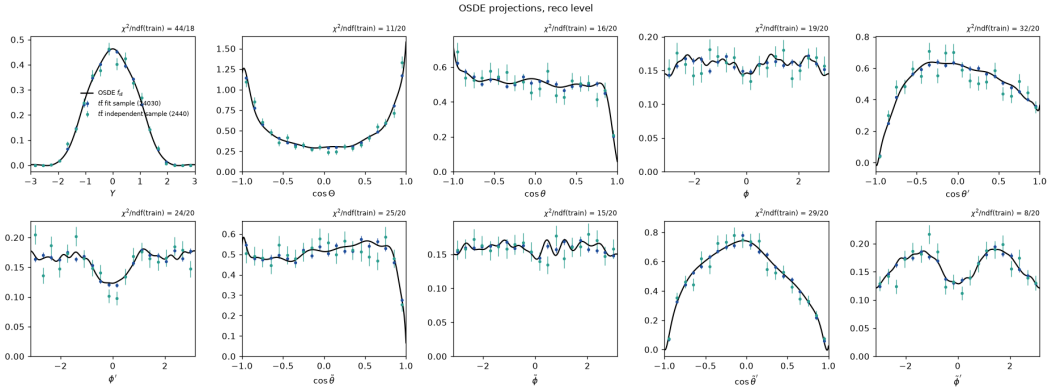
level	basis	Y	$\cos\Theta$	$\tilde{\phi}'$	$\cos\tilde{\theta}'$	$\tilde{\phi}$	$\cos\tilde{\theta}$	ϕ'	$\cos\theta'$	ϕ	$\cos\theta$
parton	paper	28.8	7.5	1.7	0.6	1.4	1.4	1.1	1.1	0.8	1.4
	paper, CV	28.6	7.5	1.8	0.6	1.4	1.4	1.3	1.1	1.0	1.4
	extended	0.5	0.5	1.5	0.4	1.1	1.2	0.9	0.8	0.6	1.2
	extended, CV	0.5	0.7	1.6	0.4	1.2	1.2	1.1	0.8	0.8	1.2
gen	paper	1057.0	4.5	32.4	37.5	8.0	44.7	29.7	40.9	5.3	68.3
	paper, CV	1127.1	4.6	32.5	37.5	8.2	44.7	30.1	40.9	5.3	68.3
	extended	16.1	1.3	0.4	0.5	0.2	0.2	0.2	0.4	0.1	0.7
	extended, CV	39.4	1.5	0.6	0.5	0.8	0.5	0.7	0.3	0.3	0.8
reco	paper	473.8	3.4	22.1	24.6	2.1	25.8	9.1	66.9	2.9	31.2
	paper, CV	473.3	3.4	22.1	24.2	2.2	26.6	9.1	66.9	2.9	31.2
	extended	9.6	0.4	0.3	0.7	0.2	1.1	0.5	0.4	0.1	0.6
	extended, CV	2.5	0.6	0.4	1.4	0.7	1.3	1.2	1.6	0.9	0.8

Extended basis (after CV truncation): χ^2/ndf $\approx$ 0.4–2 for every angle at every level; the paper basis is good only at parton level (apart from Y, $\cos\Theta$).

Extended $f_{t\bar{t}}^{\text{gen}}$ (CV-truncated): projections



Extended $f_{t\bar{t}}^{\text{reco}}$ (CV-truncated): projections

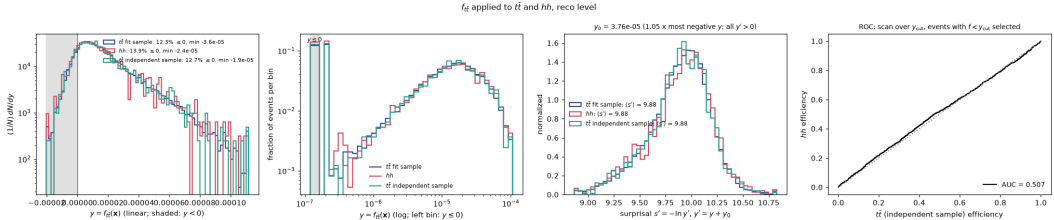


Extended basis: performance

level	terms	number	AUC (hh vs $t\bar{t}$)	$f \leq 0$ ($t\bar{t}$ indep.)	$f \leq 0$ (hh)
parton	all terms	14436	0.655	1.4%	5.8%
	CV best	333	0.666	0.3%	2.5%
	5σ	57	0.669	0.1%	2.3%
gen	all terms	32494	0.528	5.7%	8.4%
	CV best	679	0.535	1.6%	2.4%
	5σ	185	0.537	1.5%	2.2%
reco	all terms	49867	0.507	12.7%	13.9%
	CV best	323	0.512	1.4%	1.8%
	5σ	139	0.519	1.6%	1.7%

A much better description of $t\bar{t}$ hardly changes the hh separation at gen/reco: the limitation is the $t\bar{t}$-hypothesis reconstruction, not the density. Without truncation the many high-order terms add noise ($f \leq 0$ up to 13%).

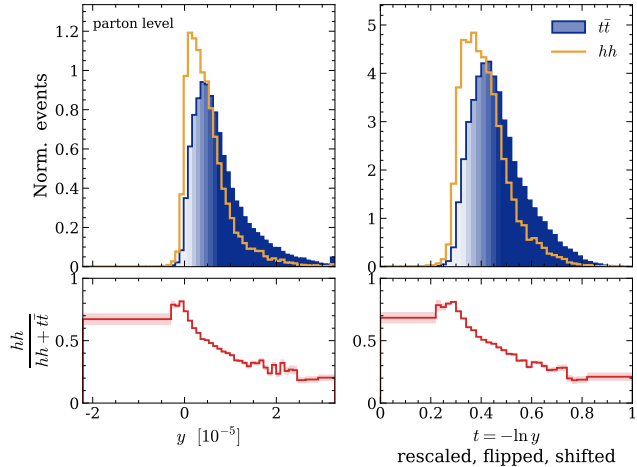
Extended $f_{t\bar{t}}^{\text{reco}}$ applied to reco-level $t\bar{t}$ and hh



Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

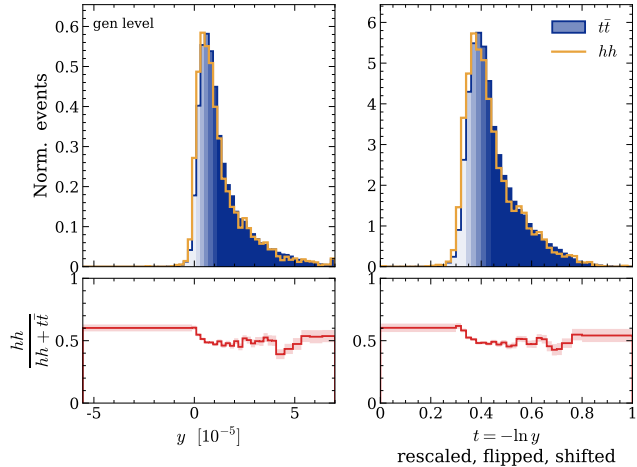
$y = f_{t\bar{t}}^{\text{parton}}$ and t, parton level (paper basis)



$y' = y - y_0$, $y_0 = 2.5 \times \min y$ (inferred from the paper figure); $s' = -\ln y'$; $t = (s'_1 - s')/(s'_1 - s'_0)$.

$t\bar{t}$: fit sample (leave-one-out); unit area.

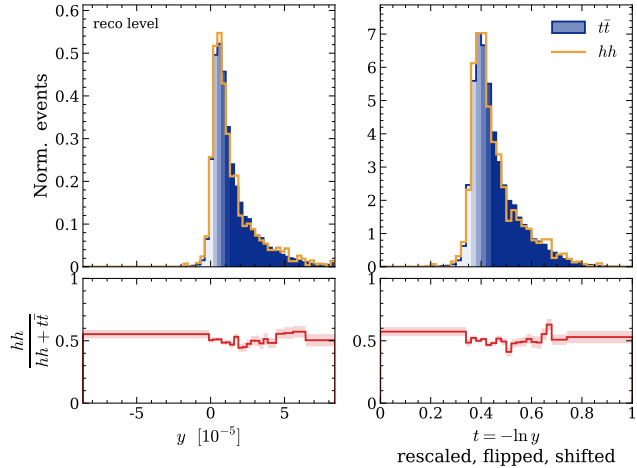
$y = f_{t\bar{t}}^{\text{gen}}$ and t, gen level (paper basis)



$y' = y - y_0$, $y_0 = 2.5 \times \min y$ (inferred from the paper figure); $s' = -\ln y'$; $t = (s'_1 - s')/(s'_1 - s'_0)$.

$t\bar{t}$: fit sample (leave-one-out); unit area.

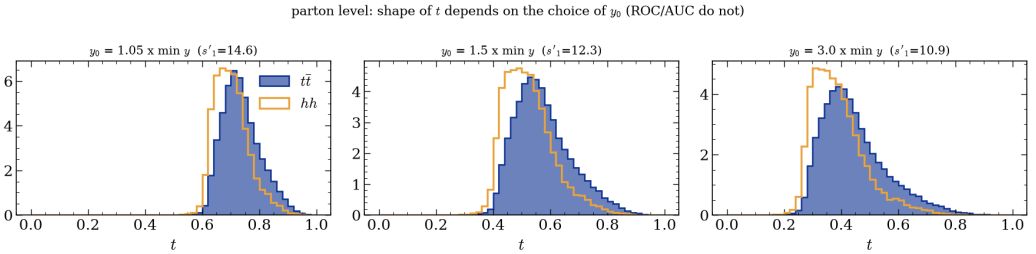
$y = f_{t\bar{t}}^{\text{reco}}$ and t, reco level (paper basis)



$y' = y - y_0$, $y_0 = 2.5 \times \min y$ (inferred from the paper figure); $s' = -\ln y'$; $t = (s'_1 - s')/(s'_1 - s'_0)$.

$t\bar{t}$: fit sample (leave-one-out); unit area.

Shape of t depends on y_0



s_1' is set by the most negative event, so y_0 just below $\min y$ squeezes t toward 1. The paper figure (axis from -1.5×10^{-5}, t populated from 0) implies $y_0 \approx 2.5 \times \min y$, used here; please confirm. The ROC/AUC do not depend on y_0 (monotonic).

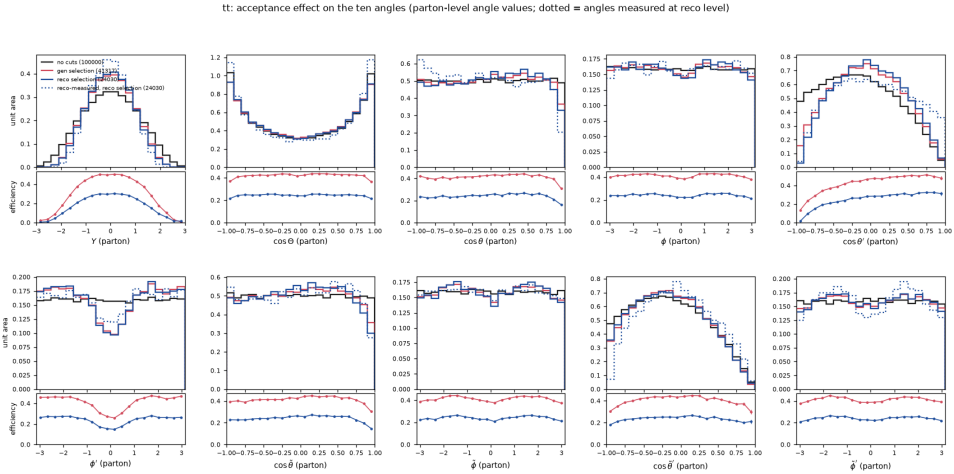
Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

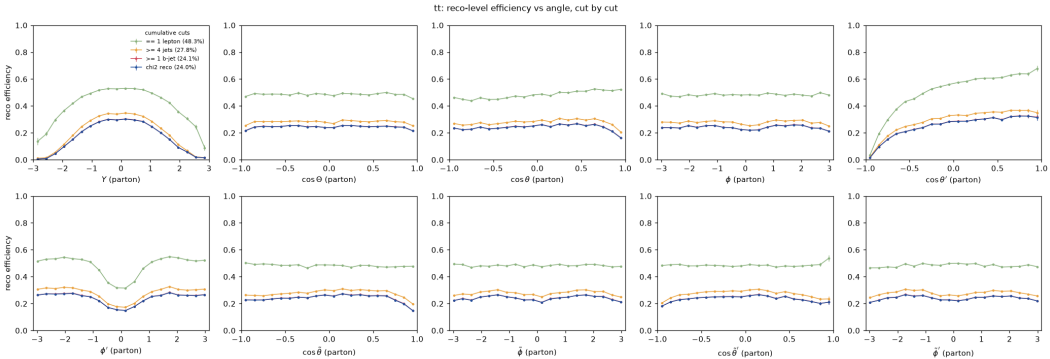
Acceptance: how the selection sculpts the angles

- Same (parton-level) angle values for every curve, so shape changes are acceptance only; dotted = angles measured at reco level (adds resolution and jet assignment)
- $\cos\theta' \to -1$, $\phi' \approx 0$: the lepton is emitted against the top's flight direction and fails $p_T > 25$ GeV (already at gen level, so not isolation)
- $|Y| \gtrsim 2$: $|\eta| < 2.5$ acceptance. $\cos\theta, \cos\tilde{\theta} \to +1$: the b quark goes backward and is too soft for the 25 GeV jet cut
- $\cos\Theta$ and the ϕ angles: nearly flat efficiency
- hh: the $t\bar{t}$-hypothesis χ^2 turns flat parton $\cos\Theta$ into $t\bar{t}$-like peaks at ± 1 (dotted): the reconstruction, more than the acceptance, makes hh look like $t\bar{t}$

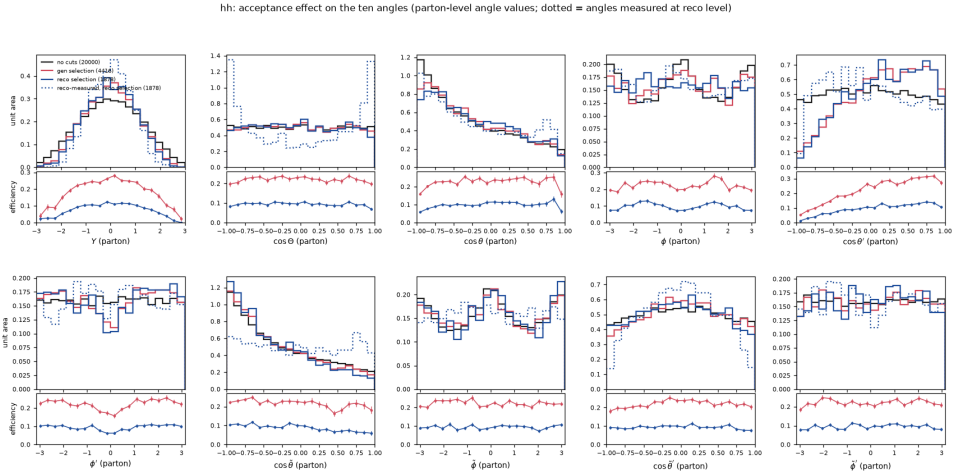
Acceptance, $t\bar{t}$: no cuts vs after cuts



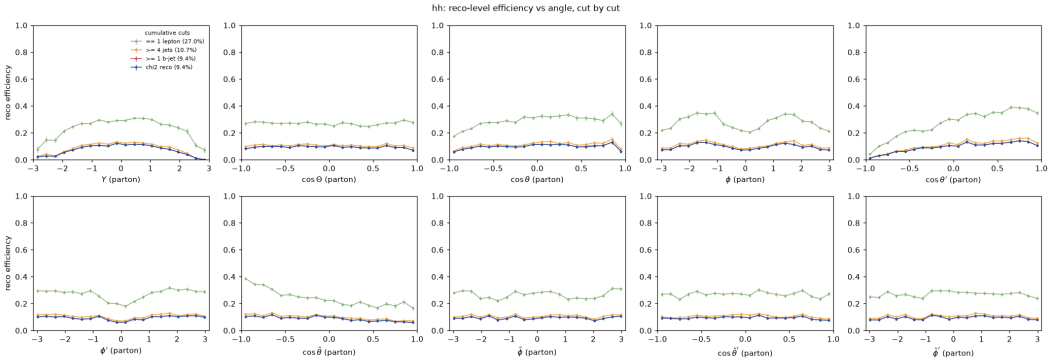
Acceptance, $t\bar{t}$: efficiency vs angle, cut by cut



Acceptance, hh: no cuts vs after cuts



Acceptance, hh: efficiency vs angle, cut by cut



Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Discriminator t (paper style)
9. Acceptance
10. Summary

Summary (1/3): samples and angles

- $t\bar{t}$ samples (parton, gen, reco; ATLAS card, no pileup) produced and validated; hh (HEFT) likewise
- Ten angles as in the paper (diagram); parton level reproduces the SM W helicity fractions ($F_0 = 0.715 \pm 0.013$)
- Reco angles track parton angles when the jets are assigned correctly; combinatorics dominate the smearing
- ν p_z: 54 / 72 GeV (gen / reco, 68% half-width); the top-mass choice of the solution is not better than smaller $|p_z|$ (63% vs 65% closer to truth)

Summary (2/3): $f_{t\bar{t}}$ and hh

- $f_{t\bar{t}}$ from 14400 OSDE coefficients; AUC(hh vs independent $t\bar{t}$): parton 0.65, gen 0.53, reco 0.51
- Separation is lost at gen/reco because the $t\bar{t}$-hypothesis χ^2 makes hh look like $t\bar{t}$: e.g. a hadronic W^* (true median 32 GeV) is reconstructed near m_W; acceptance sculpts $\cos\theta'$, ϕ', Y
- Fit-sample vs independent-sample differences are entirely the self-inclusion bias (removed by leave-one-out)
- Only $\sim$50–350 coefficients matter: CV keeps 326, 459, 342 (parton, gen, reco); simplified f has fewer negative values and the same AUC; parton level resolves $t\bar{t}$ spin correlations

Summary (3/3): $f_{t\bar{t}}$ and hh

- Paper basis has the right order at parton level, but gen/reco acceptance needs $j_{1,2}$ up to 13–16: the extended block basis (333, 679, 323 terms after CV) describes every angle (χ^2/ndf $\approx$ 0.3–1.6; gen Y still imperfect); the hh AUC is unchanged
- Paper-style t: its shape depends on the choice of y_0 (AUC does not)
- Next: equiprobable-contour binning; truncated/extended f as default?; y_0 convention; 8-angle set; an f_{hh} for A–B testing; pileup